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### 2.5.4 Selaginella

Division – Lycophyta
Class – Ligulopsida
Order – Selaginellales
Family –Selaginellaceae
Genus – *Selaginella*

*Selaginella* is commonly called 'spike moss'. They are distributed in humid temperate and tropical rain forests. *Selaginella rupestris* and *Selaginella lepidophylla* are Xerophytic. *Selaginella kraussiana*, *Selaginella chrysocaulos, Selaginella megaphylla* are some common species. In few *Selaginella* species during dry season the entire plant body gets curled and become fresh, green when moisture is available. Due to this they are called **Resurrection plants**. Example *S. lepidophylla*

**External morphology**

**Habit**

The plant body of *Selaginella* is sporophyte (2n) and it is differentiated into root, stem, and leaves (Figure 2.25). There exist variations in the habit of *Selaginella*. Some species possess prostrate creeping system (*S. kraussiana*); suberect (*S. rupestris*); erect (*Selaginella erythropus*); Climbing (*Selaginella alligans*). *S. oregana* is an epiphyte. Most of the species are perennials. on the basis of structure of stem and arrangement of leaves, *Selaginella* is divided into two sub genera namely Homoeophyllum and Heterophyllum.

Homeophyllum include species with erect stem and spirally arranged leaves. (Example: *S. upestris* and *S. oregana*). Heterophyllum include species with prosrate stem with short erect branches and dimorphic leaves (Example: *S. kraussiana* and *S. lepidophylla*).

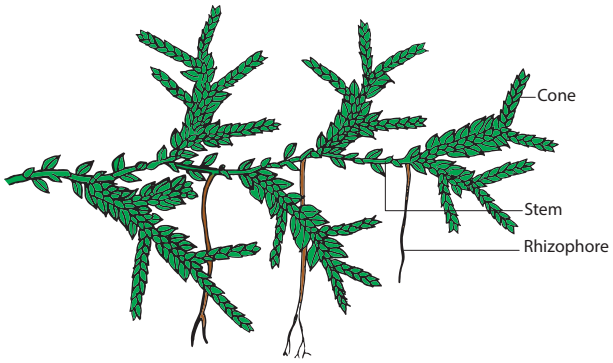



**Figure 2.25:** *Selaginella* Habit

**Root**

Primary roots are short lived and the adult plant has adventitious roots. The root may arise at the point of dichotomous branching or knot like swelling present at the basal portion of the stem. Roots are endogenous in origin.

**Rhizophore**

In many species long, cylindrical, unbranched and leafless structures arise from the lower side of the stem at the point of dichotomy called rhizophores. They grow vertically downwards and produce tufts of adventitious roots.

**Stem**

The stem may be erect, dichotomously branched or prostrate with lateral branching. The prostrate stem is dorsiventral.

**Leaves**

The leaves are microphyllous, sessile and simple. A single midvein is present in the leaves. The vegetative leaf as well as the sporophyll bears a small membranous tongue like structure on adaxial surface called **ligule**. The basal part of the ligule possess a hemispherical mass of thin walled cells called **glossopodium**. The function of ligule is not known,